FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 2.2: CHEMICAL FAMILIES	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You will write a structured, evidence-based response to the driving question for this unit. Your answer should show everything you have learned across all 8 lessons — connecting the anchoring phenomenon (the orange glow of Thika Road street lamps, the inert welding shield gas, the fizzing water treatment tablet, and the harmless party balloon) to the big idea of chemical families. Read each section prompt carefully, then write your response using evidence, scientific vocabulary, and real examples from Kenya and your daily life. Your explanation should tell a complete story: from the patterns you noticed, to the structure that explains them, to the real-world applications that depend on those patterns. Write in full sentences and paragraphs. Diagrams or electron-dot models may be included where they strengthen your explanation.

Section 1 — The Phenomenon and the Pattern: Why Do Elements Behave So Differently?	
Return to the anchoring phenomenon. Describe at least THREE contrasting behaviours you observed or studied during this unit — for example, the orange glow of sodium street lamps on Thika Road, the complete inertness of argon in a welder's torch at Kamukunji, and the sharp fizzing of calcium hypochlorite in a water tank. For each behaviour, explain what the element is doing and why it is surprising or puzzling. Then state the central pattern you discovered: elements can be sorted into 'families' based on similar properties. What evidence from the periodic	During this unit I noticed three striking contrasts in element behaviour. First, sodium — the element inside the tall street lamps along Thika Road — glows a vivid orange-yellow when electricity excites its atoms, yet a small piece of pure sodium dropped into water reacts violently, fizzing and sometimes catching fire. Second, argon — the invisible gas that surrounds a jua kali welder's flame in Kamukunji — does absolutely nothing: it neither burns, nor reacts with the hot metal, nor changes colour. Third, chlorine compounds in the white tablets a parent drops into the household water storage drum fizz immediately, release a sharp choking smell, and kill bacteria — showing chlorine is extremely reactive. These three behaviours are surprising because all three substances are made of atoms and are listed on the same periodic table, yet they behave as if they belong to completely different worlds. The pattern I discovered is that elements in the same vertical column (group) of the periodic table share similar chemical properties because they have the same number of valence electrons. Sodium sits in Group 1, alongside lithium and potassium — all of which react vigorously with water. Argon sits in Group 18 (the Noble Gases), alongside helium and neon — all of which are chemically inert. Chlorine sits in Group 17 (the Halogens), alongside fluorine and bromine — all of which are reactive non-metals that form salts. The periodic table is therefore not just a list; it is a map of chemical families.

table supports the idea that family membership — not random chance — explains these differences?	

Section 2 — Atomic Structure Explains Family Behaviour: Valence Electrons and the Periodic Table	
Explain HOW the internal structure of an atom — specifically the number and arrangement of electrons in shells — determines which chemical family an element belongs to and how it will behave. In your answer: (a) define valence electrons and explain why they matter; (b) draw or describe the electron shell diagrams for at least ONE element from THREE different groups (e.g. sodium from Group 1, chlorine from Group 17, and argon from Group 18); (c) explain why elements in the same group have similar reactivity; and (d) describe the trend in reactivity as you move DOWN Group 1 (alkali metals) and DOWN Group 17 (halogens), using specific elements as examples.	An atom is made of a nucleus containing protons and neutrons, surrounded by electrons arranged in shells. The electrons in the outermost shell are called valence electrons, and they are the key to chemical behaviour — they are the electrons that interact when atoms bond or react. Sodium (Na, atomic number 11) has the electron arrangement 2, 8, 1. Its single valence electron is loosely held and easily given away, which is why sodium reacts so violently with water — it donates that electron almost instantly. Chlorine (Cl, atomic number 17) has the arrangement 2, 8, 7. With seven valence electrons, it needs just one more to complete its outer shell, making it extremely eager to gain an electron — which is why chlorine compounds are powerfully reactive and why chlorine bleaches and disinfects. Argon (Ar, atomic number 18) has the arrangement 2, 8, 8. Its outer shell is completely full; it has nothing to give and no space to receive, so it does not react at all — which is exactly why it is used as a shield gas in welding and why helium in a party balloon is perfectly safe to breathe near. Elements in the same group share the same number of valence electrons, which is why they share similar chemical properties — this is the definition of a chemical family. Reactivity in Group 1 increases as you go down the group: lithium reacts gently with water, sodium reacts more vigorously, and potassium reacts so fast it bursts into a lilac flame. This is because each successive element has its valence electron in a shell that is farther from the nucleus, held less tightly, and therefore easier to lose. In Group 17, reactivity decreases going down: fluorine is the most reactive halogen, chlorine is next, then bromine, then iodine — because as the atom gets larger, the incoming electron is farther from the attractive nucleus and the gain is less energetically favourable.

Section 3 — Chemical Families in Kenyan Daily Life: Connecting Properties to Applications	
Choose THREE chemical families studied in this unit (for example: Alkali Metals, Halogens, Noble Gases, Transition Metals, or Alkaline Earth Metals). For each family, describe: (a) the key physical and chemical properties of the family; (b) at least ONE specific element from that family; and	Family 1 — Noble Gases (Group 18): Noble gases have full outer electron shells, which means they do not react with any other substance. They are colourless, odourless, and chemically inert. Argon, a noble gas, is used by jua kali welders in Kamukunji as a 'shield gas' — it surrounds the welding arc and prevents the extremely hot molten metal from reacting with oxygen or nitrogen in the air, which would weaken the weld. Without argon's inertness, every weld on a Nairobi metal gate or window grille would be full of brittle oxidised spots. Neon and sodium vapour (a related concept) produce coloured glows in discharge tubes, seen in Nairobi city signage. Family 2 — Halogens (Group 17): Halogens have seven valence electrons and are highly reactive non-metals. They readily gain one electron to form negative ions (halide ions). Chlorine is used in water treatment across Kenya — for example, the white calcium hypochlorite tablets dropped into household water storage tanks in Mombasa and Nairobi kill bacteria and make water safe to drink. Chlorine's high reactivity is the very property that makes it lethal to microorganisms. Iodine, another halogen, is added to table salt (iodised salt, widely sold in Kenyan shops) to prevent iodine-deficiency disorders — a direct public health

(c) a specific, real Kenyan application that DEPENDS on those properties. Your examples must be drawn from Kenyan contexts — think about farming in the Rift Valley, street lighting in Nairobi, water treatment in Mombasa, electrical wiring in homes, roofing materials (mabati), jua kali welding, or agricultural tools like the jembe.	application of halogen chemistry. Family 3 — Transition Metals (Groups 3–12): Transition metals are hard, strong, shiny solids with high melting points that conduct electricity and heat well. They often form coloured compounds and can act as catalysts. Iron is the most important transition metal in Kenya: it forms the steel used in jembe blades that till farms in the Rift Valley, in the mabati (corrugated iron sheets) that roof millions of Kenyan homes, and in the reinforcing bars inside concrete buildings in Nairobi. Copper, another transition metal, is used extensively in electrical wiring throughout Kenya because it conducts electricity with very low resistance. Zinc is electroplated onto iron (galvanisation) to protect mabati roofs from rusting — zinc's reactivity means it corrodes preferentially, sacrificing itself to protect the iron beneath, a process farmers in Kericho and Kisumu depend on to keep their rooftops intact for decades.

Section 4 — Answering the Driving Question: How Do Family Properties Explain the Materials Around Us?

Now write a direct, evidence-rich answer to the driving question: 'Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm — and how do their "family" properties explain the materials that build, light up, and sustain our daily lives?' Your answer must: (a) link each type of behaviour (explosive, inert/glowing, rusting/corroding) to a specific chemical family and its electron configuration; (b) explain how engineers, farmers, and manufacturers in Kenya deliberately CHOOSE elements based on family properties; and (c) argue why understanding chemical families is essential knowledge — not just for scientists, but for any Kenyan citizen making decisions about water safety, building materials, food preservation, or energy.	Some elements explode in water because they belong to the Alkali Metal family (Group 1), where every member has exactly one valence electron that it gives away with almost no resistance. Sodium, potassium, and lithium are all in this family — and all react violently with water, releasing hydrogen gas and heat. The further down the group, the more explosive the reaction, because the valence electron is held less tightly. By contrast, elements like argon, neon, and helium glow silently in sealed glass tubes when electricity passes through them because they belong to the Noble Gas family (Group 18). Their completely full outer shells mean they cannot form chemical bonds or react — but electricity can excite their electrons to higher energy levels, and when those electrons fall back down they release energy as coloured light. This is why the street lamps on Thika Road glow orange (sodium vapour) and why neon signs glow red in Nairobi shops. Iron rusts on a Kenyan farm because it belongs to the Transition Metals, which readily lose electrons to form positive ions when exposed to oxygen and water — a process called oxidation or corrosion. The iron in a farmer's jembe in the Rift Valley reacts slowly with rainwater and air to form iron oxide (rust), weakening the blade over time. Kenyan engineers and manufacturers use their knowledge of chemical families to make deliberate, life-improving choices. Electricians choose copper wire because the Transition Metal family offers high conductivity and moderate reactivity — copper does not corrode quickly in normal conditions. Hardware suppliers coat iron roofing sheets with zinc (galvanisation) because zinc's position in the Transition Metals means it is more reactive than iron and will corrode first, protecting the roof underneath. Public health officers choose chlorine-based tablets to treat water in Mombasa because halogens' high reactivity makes them lethal to pathogens. Understanding chemical families is therefore not an abstract school exercise — it is the knowledge behind every safe water supply, every building that does not collapse, every electrical wire that does not overheat, and every roof that does not rust away in the first rainy season. For any Kenyan citizen — farmer, builder, parent, or engineer — knowing which family an element belongs to means knowing whether it is safe, useful, or dangerous in a given situation.

Section 5 — Your Updated Model: Drawing the Big Picture

In this final section, present your updated conceptual model of chemical families. Your model must show: (a) how atomic structure (number of protons, electron shells, valence electrons) determines an element's group and period on the periodic table; (b) how group membership predicts chemical and physical properties; (c) at least FOUR elements from at least THREE different groups, with their electron configurations and one real Kenyan application each; and (d) one remaining question you still have about chemical families, elements, or materials — something this unit made you curious about that you would like to investigate further. You may present this section as a labelled diagram, a table, a written explanation, or a combination of all three.	My updated model of chemical families can be summarised in three connected ideas: Structure → Family → Application. First, every element has a fixed number of protons (its atomic number) and a specific arrangement of electrons in shells. The number of valence electrons — electrons in the outermost shell — places the element in a group (vertical column) of the periodic table. Elements in the same group form a chemical family because they share the same number of valence electrons and therefore react in similar ways. Second, family membership reliably predicts behaviour: Group 1 elements (1 valence electron) are reactive metals that lose electrons easily; Group 17 elements (7 valence electrons) are reactive non-metals that gain electrons easily; Group 18 elements (8 valence electrons, or 2 for helium) are inert; Transition Metals are hard, conductive, and moderately reactive. Third, these predictable properties are the reason specific elements are chosen for specific Kenyan applications: Sodium (Group 1, 2,8,1) — used in street lamp discharge tubes on Thika Road; its excited electrons emit orange-yellow light. Chlorine (Group 17, 2,8,7) — used in water purification tablets in Kenyan homes and municipal water supplies; its high reactivity kills bacteria. Argon (Group 18, 2,8,8) — used as a welding shield gas in jua kali workshops; its inertness protects hot metal from oxidation. Iron (Transition Metal, 2,8,8,2 + d-electrons) — used in jembe blades and mabati roofing across Kenya; its strength and moderate reactivity make it useful, though zinc coating is needed to slow corrosion. Copper (Transition Metal) — used in electrical wiring nationwide; its excellent conductivity and low indoor-corrosion rate make it ideal. One question I still have: If noble gases are so completely inert, how exactly does electricity make them glow — does the electron temporarily gain energy and release it as light without ever actually bonding with another atom? I would like to investigate atomic emission spectra further to understand how different elements produce different colours of light, and whether this principle could be used to identify unknown elements found in Kenyan soil or water.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Phenomenon Connection	The response consistently and explicitly connects all major explanations back to the anchoring phenomenon (Thika Road street lamps, Kamukunji welder's argon, water treatment tablets, party balloon helium). The student uses phenomenon details as evidence throughout every section, not just in the introduction.	The response connects most explanations to the anchoring phenomenon. At least three of the four phenomenon elements (sodium glow, argon inertness, chlorine fizzing, helium balloon) are referenced with some scientific detail.	The response mentions the anchoring phenomenon in passing (e.g., 'the street lamps on Thika Road') but does not use it as evidence or return to it when making scientific arguments. The connection feels superficial.
Atomic Structure and Valence Electrons	The student accurately explains electron shell configurations for at least three elements from different groups, correctly uses the term 'valence electrons,' and clearly explains the causal link between valence electron count, group	The student correctly describes electron arrangements for at least two elements and explains that valence electrons determine group membership and reactivity. One directional trend (e.g., reactivity down Group 1) is mentioned	The student demonstrates partial understanding of electron shells but conflates periods and groups, or incorrectly states electron configurations. The link between valence electrons and reactivity is stated but not explained

	membership, and chemical reactivity — including directional trends (reactivity increases/decreases down a group) with specific examples.	with at least one example, though the explanation may lack full depth.	causally (e.g., 'sodium reacts because it is in Group 1' without explaining why Group 1 elements react).
Chemical Families — Properties and Applications	The student describes at least three chemical families with accurate physical and chemical properties, names specific elements from each family, and links each to a precise, real Kenyan application (e.g., argon in Kamukunji welding, chlorine in Mombasa water treatment, zinc galvanisation on mabati roofs). The connection between the property and the application is explicitly stated (e.g., 'argon is used because its inertness prevents oxidation').	The student describes at least two chemical families accurately and provides a Kenyan application for each. The connection between property and application is stated for most examples, though one application may be generic (e.g., 'copper is used in wires') without explaining which property makes it suitable.	The student names chemical families and lists some properties but struggles to connect properties to applications. Applications may be non-Kenyan, vague, or disconnected from the properties described (e.g., listing that chlorine is reactive but not explaining how that reactivity makes it useful for water treatment).
Driving Question — Depth and Accuracy of Final Answer	The student's answer to the driving question is coherent, evidence-rich, and explicitly addresses all three types of element behaviour (explosive/reactive, inert/glowing, corroding/rusting) by linking each to a specific chemical family and electron configuration. The student argues convincingly and with specific Kenyan examples why this knowledge matters for citizens, not just scientists.	The student addresses at least two of the three behaviour types (reactive, inert, corroding) and links them to chemical families. The argument about real-world relevance is present but may be general rather than grounded in specific Kenyan citizen decisions (farming, building, water safety, etc.).	The student attempts to answer the driving question but the response is incomplete — only one behaviour type is addressed, or the link between electron structure, family membership, and behaviour is missing. The real-world relevance section is absent or limited to a single vague sentence.
Scientific Communication and Use of Vocabulary	The student consistently and correctly uses key scientific vocabulary throughout: valence electrons, electron configuration, group, period, reactivity, inert, noble gas, halogen, alkali metal, transition metal, oxidation, ion, electron shell. Explanations are clear, logically sequenced, and accessible. Diagrams or tables (if included) are labelled and directly support the written argument.	The student uses most key vocabulary correctly in context. Occasional misuse or omission of terms (e.g., using 'layer' instead of 'shell,' or omitting 'valence') does not significantly obscure meaning. The explanation is generally clear and follows a logical sequence.	The student uses limited scientific vocabulary or uses terms incorrectly (e.g., confusing 'group' and 'period,' or using 'charge' when meaning 'valence electrons'). Explanations are hard to follow in places, and the response may read more like a list of facts than a connected scientific argument.